

Data Structures



Submitted to

Mr. Bilal Bukhari

Submitted by

Laraib Khalid

(fa25-bcs-059)

COMSATS University Islamabad,

Attock Camous.

Lab Activity 02 (Pointers)

Code:

```
#include <iostream>

using namespace std;

int main() {

    int var = 10;

    // declare pointer and store address of var

    int* ptr = &var;

    // print value and address

    cout << "Value of var: " << var << endl;

    cout << "Address of var: " << &var << endl;

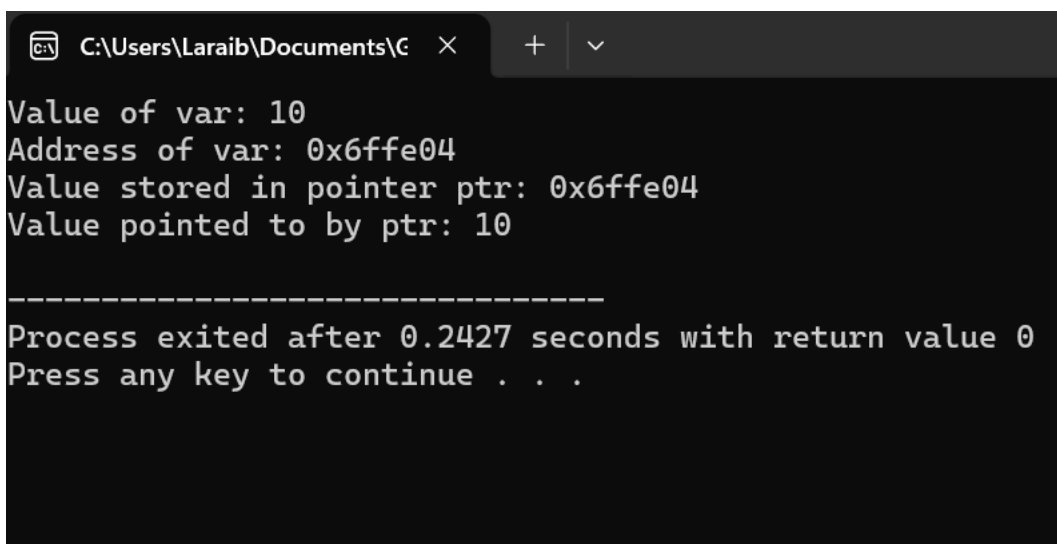
    cout << "Value stored in pointer ptr: " << ptr << endl;

    cout << "Value pointed to by ptr: " << *ptr << endl;

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\G  X  +  v

Value of var: 10
Address of var: 0x6ffe04
Value stored in pointer ptr: 0x6ffe04
Value pointed to by ptr: 10

-----
Process exited after 0.2427 seconds with return value 0
Press any key to continue . . .
```

Code:

```
#include <iostream>

using namespace std;

int main() {

    int var = 10;

    // Store the address of

    // var variable

    int* ptr = &var;

    // Access value using (*)

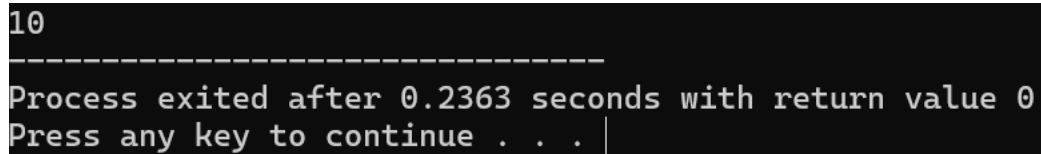
    // operator

    cout << *ptr;

    return 0;

}
```

Result:

A screenshot of a terminal window with a black background and white text. The first line shows the output '10'. The second line is a horizontal dashed line. The third line says 'Process exited after 0.2363 seconds with return value 0'. The fourth line says 'Press any key to continue . . . |' with a vertical bar at the end.

```
10
-----
Process exited after 0.2363 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int main() {

    int var = 10;

    // Store the address of

    // var variable

    int* ptr = &var;

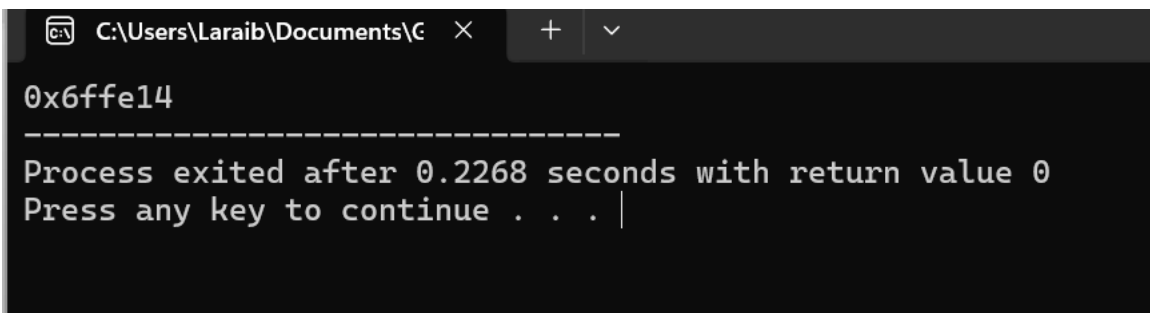
    // Access the address value

    cout << ptr;

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\C >  +  v

0x6ffe14
-----
Process exited after 0.2268 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int main() {

    int a = 10;

    int b = 99;

    int *ptr = &a;

    cout << *ptr << endl;

    // Changing the address stored

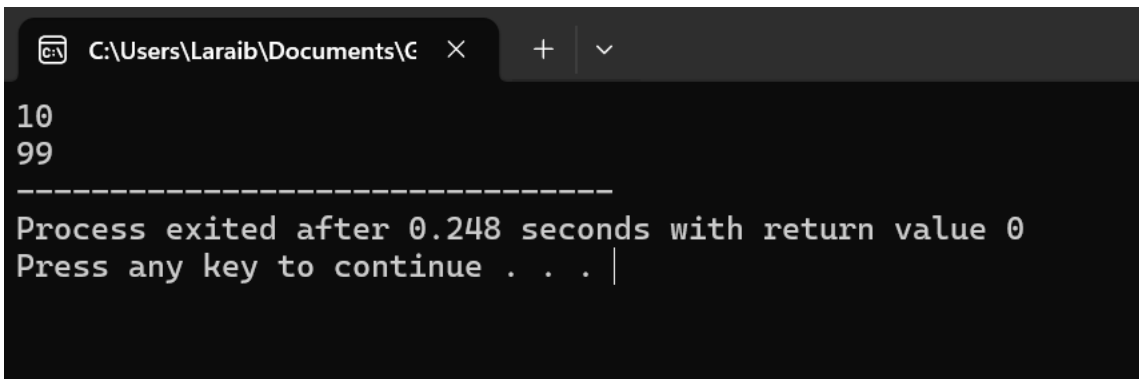
    ptr = &b;

    cout << *ptr;

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\C  ×  +  ∨

10
99
-----
Process exited after 0.248 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int main() {

    int *ptr1;

    char *ptr2;

    // Finding size using sizeof()

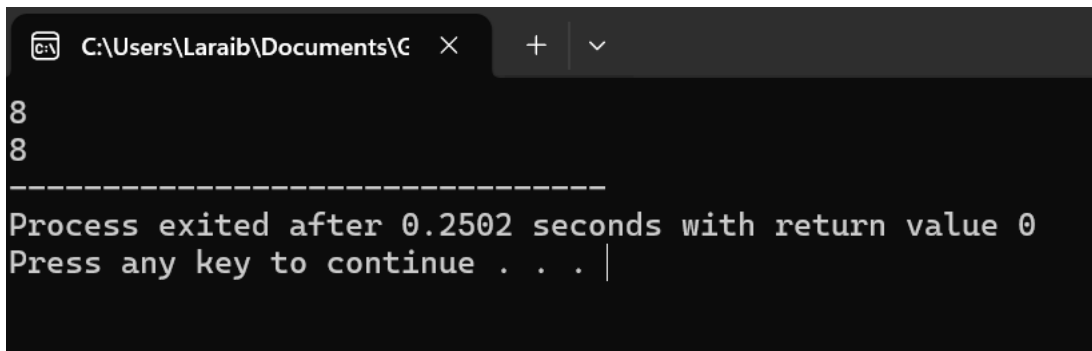
    cout << sizeof(ptr1) << endl;

    cout << sizeof(ptr2);

    return 0;

}
```

Result:



The screenshot shows a Windows command prompt window with the title bar "C:\Users\Laraib\Documents\C" and standard window controls. The output of the program is displayed in a monospaced font: the number "8" is printed on two separate lines, followed by a dashed line separator. Below the separator, the text "Process exited after 0.2502 seconds with return value 0" is shown, followed by "Press any key to continue . . . |" with a vertical cursor.

```
C:\Users\Laraib\Documents\C  ×  +  ∨

8
8
-----
Process exited after 0.2502 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int main() {

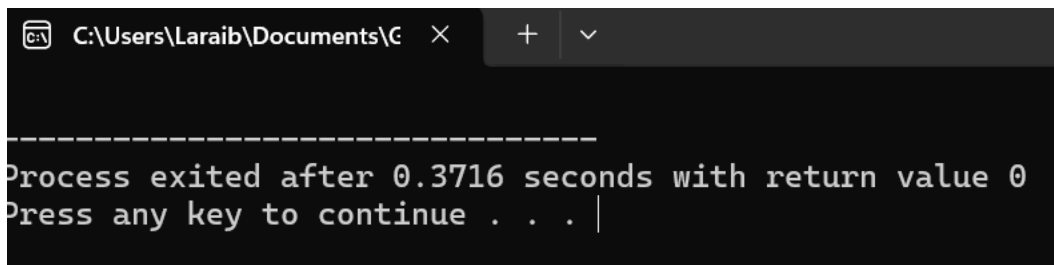
    // Wild pointer

    int *ptr;

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\G  X  +  v

-----
Process exited after 0.3716 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int main() {

    // nullptr pointer

    int *ptr = nullptr;

    return 0;

}
```

Result:

Compiler (3) Resources Compile Log Debug Find Results Close			
Line	Col	File	Message
		C:\Users\Laraib\Documents\GitHub\pointer7.cpp	In function 'int main()':
5	16	C:\Users\Laraib\Documents\GitHub\pointer7.cpp	[Error] 'nullptr' was not declared in this s
5	13	C:\Users\Laraib\Documents\GitHub\pointer7.cpp	[Error] expected '}' at end of input

Code:

```
#include <iostream>

using namespace std;

int main() {
    int x = 42;

    // void pointer holding address of an int

    void* ptr = &x;

    // Error: cannot dereference void pointer

    // cout << *ptr;

    // Typecast before dereferencing

    cout << "Value pointed by void pointer: " << *(static_cast<int*>(ptr)) << endl;

    return 0;
}
```

Result:

```
C:\Users\Laraib\Documents\G x + v
Value pointed by void pointer: 42
-----
Process exited after 0.2556 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

int* getPointer() {
    int x = 10;
    // returning address of local variable
    return &x;
}

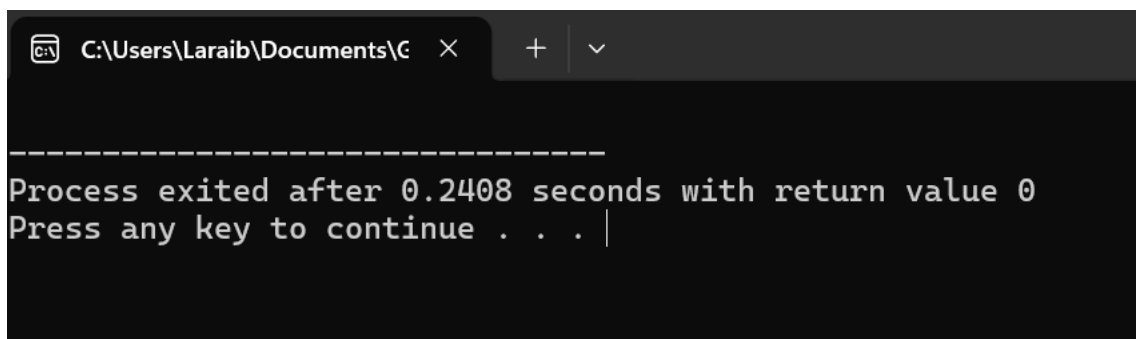
int main() {
    // ptr becomes dangling here

    int* ptr = getPointer();
    // Undefined behavior

    // cout << *ptr;

    return 0;
}
```

Result:



The screenshot shows a Windows command prompt window with a single tab titled "C:\Users\Laraib\Documents\G". The window has a dark background and displays the following text: a dashed line, "Process exited after 0.2408 seconds with return value 0", and "Press any key to continue . . . |".

Code:

```
#include <iostream>

using namespace std;

int main() {

    int var = 10;

    // Store the address of

    // var variable

    int* ptr1 = &var;

    int** ptr2 = &ptr1;

    // Access values using (*)

    // operator

    cout << *ptr1 << endl;

    cout << **ptr2;

    return 0;

}
```

Result:

```
10
10
-----
Process exited after 0.2584 seconds with return value 0
Press any key to continue . . . |
```

References

Code:

```
#include <iostream>

using namespace std;

int main() {

    int x = 10;

    // ref is a reference to x.

    int& ref = x;

    // printing value using ref

    cout << ref << endl;

    // Changing the value and printing again

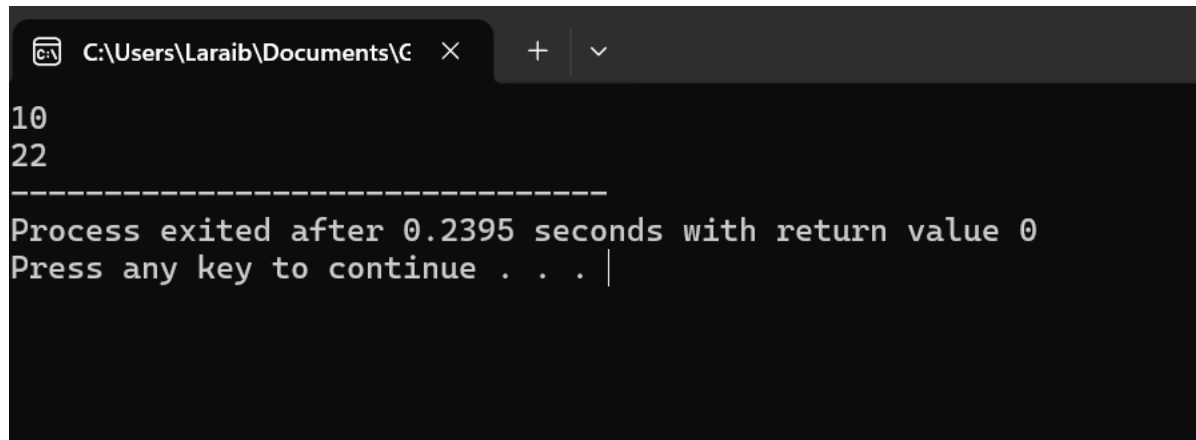
    ref = 22;

    cout << ref;

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\C  ×  +  v

10
22
-----
Process exited after 0.2395 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

using namespace std;

void modifyValue(int &x) {
    // Modifies the original variable
    x = 20;
}

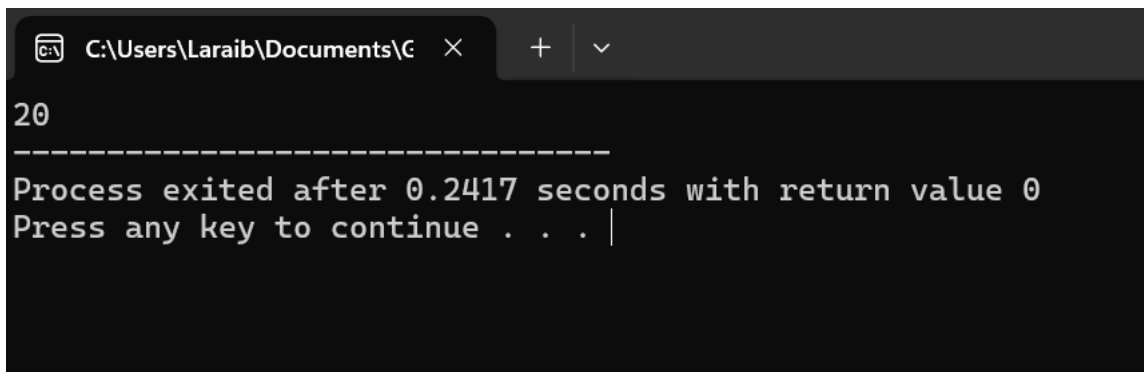
int main() {
    int a = 10;

    // Pass a by reference
    modifyValue(a);

    cout << a;

    return 0;
}
```

Result:



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'C:\Users\Laraib\Documents\G' and includes standard window controls. The output of the program is displayed in white text: the number '20' is printed on the first line, followed by a horizontal dashed line. Below the line, the text 'Process exited after 0.2417 seconds with return value 0' is shown, followed by 'Press any key to continue . . . |' where the cursor is positioned at the end of the line.

Code:

```
#include <iostream>

using namespace std;

int& getMax(int &a, int &b) {
    // Return the larger of the two numbers
    return (a > b) ? a : b;
}

int main() {
    int x = 10, y = 20;

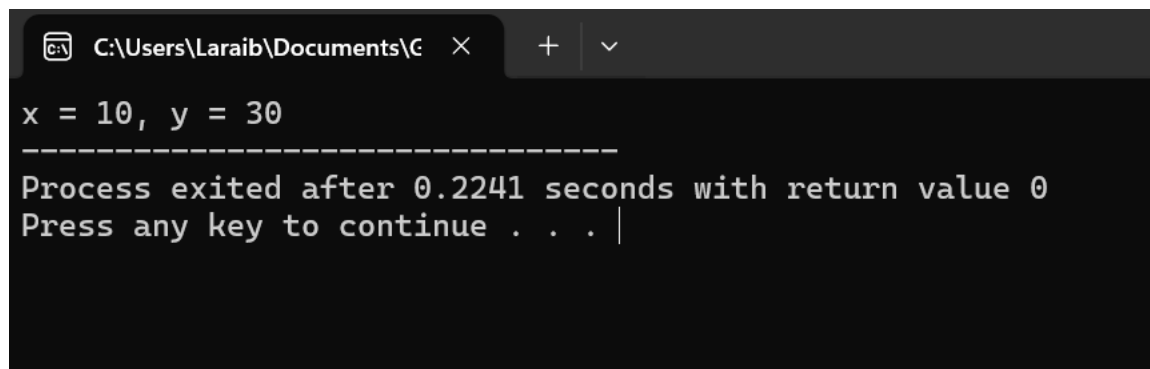
    int &maxVal = getMax(x, y);

    // Modify the value of the larger number
    maxVal = 30;

    cout << "x = " << x << ", y = " << y;

    return 0;
}
```

Result:

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\Laraib\Documents\G' and standard window controls. The command prompt displays the output of the program: 'x = 10, y = 30' followed by a dashed line separator. Below the separator, it says 'Process exited after 0.2241 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

```
C:\Users\Laraib\Documents\G  ×  +  ∨

x = 10, y = 30
-----
Process exited after 0.2241 seconds with return value 0
Press any key to continue . . . |
```

Code:

```
#include <iostream>

#include <vector>

using namespace std;

int main() {

    vector<int> vect{ 10, 20, 30, 40 };

    // We can modify elements if we

    // use reference

    for (int& x : vect) {

        x = x + 5;

    }

    // Printing elements

    for (int x : vect) {

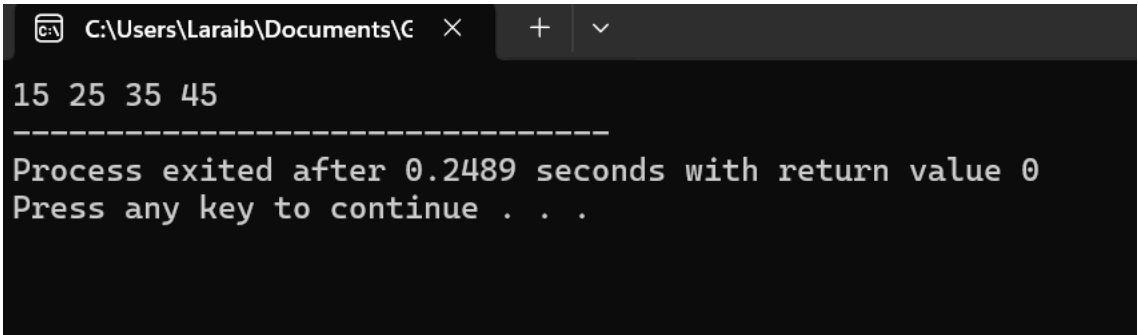
        cout << x << " ";

    }

    return 0;

}
```

Result:



```
C:\Users\Laraib\Documents\C  ×  +  ∨

15 25 35 45
-----
Process exited after 0.2489 seconds with return value 0
Press any key to continue . . .
```

